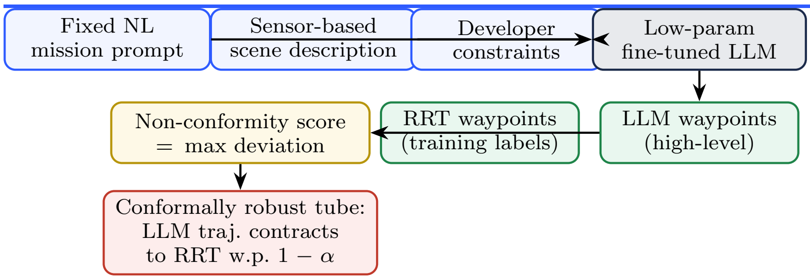
Why CP-Contracted LLM Path Planning May Not Be Justifiable

1. Current Mission



Pipeline summary: a fine-tuned LLM proposes high-level waypoints from user defined natural langauge task augmented with a scene description generated using live sensor data; RRT supplies the "ground truth" used both to generate fine-tuning data *and* to define the conformal non-conformity score. The guarantee produced is purely *geometric closeness of the LLM path to RRT* — not safety, smoothness, or task success in any independent sense.

2. Arguments Against Using LLMs for This Task

Claimed Benefit	Argument For	Argument Against	Verdict
Generalisability	LLM is a generalist by pre-training; the fine-tuning set (scene → RRT waypoints) is far smaller than what a path-planning NN would need from scratch.	The fine-tuning data is still specific to the environment and obstacle layout it was collected in — generalisability claimed at the foundation-model level is not the generalisability actually delivered at deployment.	**kind of justifiable**
Semantic Awareness	The LLM, by itself, is semantically aware of scenes, objects, and context.	This awareness is never leveraged by the formulation — worse, the CP score *penalizes* any plan that uses semantic reasoning to deviate from RRT, since deviation = non-conformity.	**Self-Defeating**
Anthropomorphic Behaviour	Pre-training on human data lets the LLM generate trajectories mimicking human intent/preference.	Any deviation from RRT is flagged as failure / non-robustness, so this inherent property is rendered redundant by the very metric meant to validate the system.	**Self-Defeating**
Human-in-the-Loop	Prompts are natural language, so a non-expert can easily modify them for everyday drone tasks.	The formulation only works for a carefully tailored prompt after iteration — performance can't be claimed for arbitrary prompts. Even granting a full agentic stack (vision encoder + separate LLM to parse user request to generate clear mission objectives form natural language inputs like "find me a pen") justifies LLMs for *task specification*, not *path planning with CP-to-RRT based robustness.*	**Not Justifiable**

- **Self-Defeating** — the benefit is real, but the CP-vs-RRT metric structurally punishes the very behaviour it claims to exploit.
- **Not Justifiable** — even granting the benefit fully, it does not argue for LLM-based *path planning* specifically.

3. Ideas Going Forward

Idea	Work Involved	Drawbacks
1. Alternative performance metrics	Inspired by arXiv:1806.04167: instead of contracting the LLM *trajectory* to RRT, contract a *property* of the plan (constraint satisfaction, stability, recursive feasibility / safety / smoothness) to nominal. This would preserve the LLM's semantic and mimicing human-intent strengths while still proving comparability to RRT on relevant metrics. Requires a new mathematical formulation based on Hoeffding's-inequality bound, with later integration of contraction theory to be assessed.	Complete departure from the current CP + contraction track — new formulation and implementation required from near scratch; feasibility of integrating contraction theory later is still unaccessed.
2. Better expert data	Keep the current CP-contraction plan; justify LLM use via generalisability, but generate expert training data that is human-supervised and semantically aware rather than pure RRT output. Requires designing a process to generate this expert trajectory data, and still defending the remaining drawbacks (semantic-awareness / anthropomorphic-behaviour rows above still apply).	Dataset generation remains an open, weakly-specified problem — by the author's own assessment, the *weaker* of the two arguments for justifying LLM use.